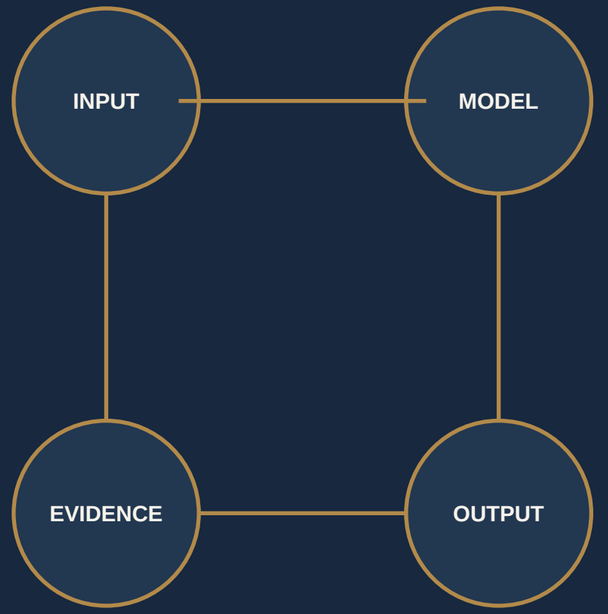
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Computer Architecture

AI.206 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define cpus.
- Apply key methods including instruction sets.
- Build or analyze a practical system involving memory hierarchy.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

CPUs

02

METHODS

Instruction sets

03

SYSTEMS

Memory hierarchy

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · CPUs
CPUs

02 · Instruction sets
Instruction sets

03 · Memory hierarchy
Memory hierarchy

04 · Buses
Buses

05 · Caching
Caching

06 · Pipelines
Pipelines

07 · GPUs
GPUs

08 · Parallelism and architecture f
Parallelism and architecture for AI workloads

Core ideas, made explicit

AI.206 • Distinguish definitions, assumptions and evidence

- CPUs
- Instruction sets
- Memory hierarchy
- Buses
- Caching

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.206 • Distinguish definitions, assumptions and evidence

- Pipelines
- GPUs
- Parallelism and architecture for AI workloads
- Mandatory practicals: Analyze processor architecture
- Benchmark systems and demonstrate how AI workloads use CPU/GPU resources

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
CPUs

INPUTS
Instruction sets

METHOD
Memory hierarchy

OUTPUT
Buses

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving cpus.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using cpus.
- Implement or simulate a baseline using instruction sets.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.